

MUDANJIANG, China

WMO: 540940

Lat: 44.5039N

Lon: 129.6683E

Elev: 307

StdP: 97.70

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.2	-24.1	-30.2	0.2	-24.6	-28.4	0.3	-22.7	10.3	-12.1	9.1	-12.3	0.9	200	0.499

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	31.5	21.3	30.0	20.7	28.6	20.3	23.7	28.6	22.8	27.2	22.0	26.2	3.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.2	17.5	26.5	21.4	16.7	25.6	20.6	15.8	24.8	72.9	28.7	69.4	27.2	66.3	26.2	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.0	6.6	-28.1	34.3	3.1	1.4	-30.3	35.3	-32.1	36.1	-33.9	36.9	-36.1	37.9		
(5)			-28.3	25.1	3.0	0.9	-30.5	25.7	-32.3	26.2	-34.0	26.7	-36.2	27.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	-15.9	-11.0	-2.5	7.2	14.4	20.2	23.0	21.7	15.5	6.8	-4.1	-13.3		
	DBStd	14.24	4.11	5.15	5.38	4.75	4.06	3.41	2.55	2.76	3.47	4.48	6.04	4.62		
	HDD10.0	3165	803	588	389	112	8	0	0	0	3	119	422	722		
	HDD18.3	5131	1061	821	647	335	133	19	2	6	98	357	672	980		
	CDD10.0	1429	0	0	1	27	145	306	402	361	166	21	1	0		
	CDD18.3	354	0	0	0	1	12	76	146	108	12	0	0	0		
	CDH23.3	2752	0	0	0	18	191	659	1083	706	92	3	0	0		
	CDH26.7	807	0	0	0	3	48	218	354	175	9	0	0	0		
(14) Wind	WSAvg	2.3	2.1	2.3	2.9	2.9	2.7	2.2	2.0	1.9	1.9	2.3	2.4	2.2		
(15) Precipitation	PrecAvg	535	4	6	10	27	49	89	118	123	56	32	15	8		
	PrecMax	748	15	18	32	77	135	204	295	348	113	98	54	30		
	PrecMin	339	0	0	1	3	6	12	20	35	7	2	0	1		
	PrecStd	104	3	5	7	18	28	45	57	73	30	22	11	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.6	5.5	15.2	25.7	30.6	33.5	33.8	32.0	27.9	22.7	14.2	3.9		
		MCWB	-4.2	0.7	6.8	13.0	16.0	19.4	22.9	23.6	19.6	14.0	7.8	1.1		
	2%	DB	-5.2	3.0	11.1	22.2	27.9	31.0	31.7	30.2	25.9	19.4	10.3	-1.2		
		MCWB	-7.4	-0.8	4.3	11.0	15.1	19.2	22.3	22.4	17.9	11.8	5.4	-3.5		
	5%	DB	-7.4	0.4	8.3	19.3	25.4	29.3	30.2	28.6	24.0	16.8	7.3	-3.9		
		MCWB	-9.1	-2.6	2.7	9.6	14.4	18.8	21.9	21.6	16.7	10.5	3.4	-5.7		
	10%	DB	-9.4	-2.1	5.8	16.4	22.8	27.2	28.7	27.1	22.3	14.4	4.6	-6.2		
		MCWB	-10.8	-4.7	1.3	7.9	13.2	18.3	21.6	20.9	15.9	9.2	1.4	-7.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-4.0	1.4	7.4	14.5	17.7	22.2	25.3	24.8	21.2	14.7	8.5	1.2		
		MCDB	-1.6	4.7	13.9	23.4	27.7	28.4	30.9	30.4	25.8	20.8	13.3	3.7		
	2%	WB	-7.1	-0.6	5.0	11.6	16.2	21.0	24.0	23.7	19.5	12.9	5.9	-3.3		
		MCDB	-5.3	2.7	10.1	20.9	25.1	27.9	29.3	28.4	23.8	17.5	9.7	-1.4		
	5%	WB	-9.1	-2.6	3.2	10.0	15.0	20.1	23.1	22.8	18.1	11.4	3.7	-5.7		
		MCDB	-7.6	0.3	7.6	18.6	23.1	26.5	27.9	26.8	22.4	15.9	7.1	-4.0		
	10%	WB	-10.9	-4.7	1.5	8.4	14.0	19.2	22.3	22.0	16.9	9.7	1.6	-7.8		
		MCDB	-9.5	-2.2	5.5	15.5	21.5	25.5	26.9	25.6	21.2	13.7	4.5	-6.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	11.8	10.6	11.9	11.7	10.8	9.2	9.3	12.0	11.7	9.6	9.3		
		MCDBR	12.1	13.3	13.7	16.8	16.8	14.1	12.2	11.4	14.0	15.3	12.9	11.1		
		MCWBR	10.3	10.1	8.3	8.4	6.7	5.0	4.6	4.7	6.5	8.3	8.7	9.3		
	5% WB	MCDBR	12.2	12.8	12.7	15.7	14.4	11.7	9.8	9.4	11.3	13.6	12.5	10.8		
		MCWBR	10.4	10.0	8.2	8.1	6.1	4.8	4.4	4.4	5.8	7.9	9.0	9.2		
(40) Clear-Sky Solar Irradiance	taub		0.326	0.366	0.452	0.547	0.583	0.572	0.540	0.492	0.440	0.463	0.388	0.340		
	taud		2.051	1.951	1.827	1.719	1.705	1.818	1.948	2.066	2.116	1.923	2.016	2.027		
	Ebn at Noon		749	783	760	723	713	725	743	765	772	667	660	678		
	Edh at Noon		104	140	182	221	231	207	180	155	135	143	109	97		
(44) All-Sky Solar Radiation	RadAvg		2.23	3.35	4.33	4.67	5.15	5.49	5.00	4.70	4.26	3.02	2.03	1.79		
	RadStd		0.19	0.17	0.27	0.36	0.46	0.59	0.45	0.41	0.34	0.23	0.16	0.11		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB		1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page